

Rahul Ghanghas

Location: Edison, New Jersey • In-office • Hybrid • Open to relocation

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PROFESSIONAL SUMMARY

Dynamic and results-driven Software Engineer with extensive experience designing and delivering high-performance, enterprise-grade applications. Skilled in architecting and implementing microservices, RESTful APIs, and modular back-end systems using Java and Spring Boot. Adept at optimizing application performance, enforcing test-driven development, and driving continuous integration and delivery in Agile environments. Passionate about writing clean, maintainable Java code that scales and meets rigorous quality standards.

TECHNICAL SKILLS

Software Development Life Cycle (SDLC) with experience in Agile methodologies.

Programming Languages: Java, JavaScript, SQL, Python

Web Technologies: HTML, CSS, React.js, Bootstrap, Node.js, Express.js, RESTful APIs

Backend: J2EE, Spring, MVC, Hibernate, Maven, Spring Boot, Spring Data JPA, JUnit, Mockito

Databases: MySQL, PostgreSQL, MongoDB, Redis

Tools: AWS, Git, Bitbucket, GitHub, Jenkins, Kafka, Docker, WorkFusion, Tableau, Linux

PROFESSIONAL EXPERIENCE

MedHarbor, Software Engineer

Jun 2024 - Present

Patient Management System: A microservices-based platform for managing patient data, appointments, and real-time notifications, integrated with dynamic dashboards for healthcare providers.

Environment: Java, Spring Boot, JUnit, Mockito, MySQL, RESTful APIs

- Designed and developed a microservices-based backend system to manage patient data, appointments, and medication details using Java, Spring Boot, and MySQL.
- Developed APIs with Spring MVC to enable seamless service-to-service communication and support integrations with external EHR platforms, lab services, and third-party applications.
- Collaborated on MySQL schema design and added targeted indexes to critical tables, improving average query response times by 30%.
- Comprehensive code coverage achieved through unit testing with JUnit and Mockito framework to ensure robust, and bug-free deployments.
- Integrated Redis via Spring Cache to cache patient lookup and appointment endpoints, reducing database load by 40% and cutting average API latency by 25%.
- Mentored junior developers on Java and Spring Boot best practices and agile development methodologies.

Bank of America, Software Engineer

Jul 2020 - Jun 2024

Email Assist: A distributed system for automating invoice processing, email communications, and real-time analytics for financial operations.

Environment: Java, Spring Boot, JUnit, Mockito, AWS, MySQL, Kafka, Jenkins, Python, WorkFusion

- Implemented automation of invoices and emails for Loans and Payments Operations, leveraging a microservices architecture to build highly available distributed systems.

- Developed scalable backend services Java, Spring Boot, RESTful APIs, SQL, and a Robotic Process Automation (RPA) tool, WorkFusion for processing 3000 Invoices per day.
- Created a new pre-processing logic to extract new fields and automate SME manual interventions, increasing data extraction accuracy by 15%.
- Utilized Kafka to efficiently manage high-volume data streams for invoice and payment processing.
- Maintained high quality code by implementing unit tests with JUnit and Mockito framework in Test-Driven development (TDD), ensuring 100% code coverage.
- Created custom Tableau widgets and dashboard elements for real-time invoice tracking, enabling dynamic visualizations and responsive data updates.
- Trained an AutoML extraction model using Python for Treasury Operations, providing 74% accuracy.
- Optimized data access and information retrieval by managing data storage on Amazon S3.
- Streamlined deployment process of 40+ Autosys jobs through CI/CD pipeline automation using Jenkins.

DocAssist: A distributed system for automating documents authentication and processing for onboarding new clients.

Environment: Java, Spring Boot, MySQL, AWS, WorkFusion, Tableau

- Developed the RV document project to streamline onboarding workflows by automating document handling and system integration using Java, Spring Boot, SQL, AWS, and WorkFusion.
- Collaborated with cross-functional teams to implement straight-through processing and automation of documents, saving 37,000 hours in capacity per year.
- Proposed and implement a new design format to enhance the performance of all the Tableau dashboards.
- Utilized Tableau to extract and analyze data. Developed interactive dashboards to facilitate effective business decisions.
- Conducted proof-of-concept to incorporate the Americans with Disabilities Act (ADA) and Web Content Accessibility Guidelines using Tableau and Power BI Server, enabling access for special needs individuals.

Inventeron Technologies, Software Engineering Intern

May 2019 - Jul 2019

Dashboard Modernization: A frontend overhaul project to replace legacy UI systems with a modular React.js component library, improving development efficiency and user experience consistency.

Environment: JavaScript, React.js, Bootstrap

- Designed a responsive web application using React.js and Bootstrap, significantly enhancing user interface aesthetics and overall user experience.
- Developed modular and reusable React.js components that reduced development time and improved consistency across the application.
- Participated in Agile sprints, contributing to daily stand-ups, code reviews, and comprehensive unit/integration testing for both front-end and back-end modules.
- Collaborated closely with senior engineers to integrate RESTful APIs into the React.js application, ensuring seamless data flow and dynamic content rendering.
- Authored detailed technical documentation for React component libraries, supporting future scalability and team collaboration.

PROJECTS

YelpCamp - Full Stack Web Application

- Developed a secure, user-friendly platform for campground reviews with a focus on robust user authentication and dynamic content rendering.
- Integrated custom React.js components for interactive search, filtering, and map-based campground visualization.
- Leveraged Node.js, Express.js, and MongoDB to create a scalable back-end infrastructure supporting real-time data updates.

- Enhanced the front end with responsive design techniques and optimized state management using React hooks and context APIs.
- Technologies: HTML5, CSS3, Bootstrap, JavaScript, React.js, Node.js, Express.js, Passport.js, MongoDB

Movie Booking Application

- Created an intuitive user interface for movie selection and ticket booking, ensuring a smooth, engaging user experience.
- Developed interactive React.js elements, such as live seat selection and booking status updates, that dynamically reflect real-time changes.
- Containerized microservices using Docker for scalable deployment and streamlined development processes.
- Implemented robust error handling and unit tests to ensure a high-quality, reliable application environment.
- Technologies: JavaScript, Bootstrap, React.js, Node.js, Express.js, MongoDB, JSON, RESTful APIs

Self-Adaptive Appearance Based Eye Tracking

- Developed a machine learning model for real-time eye tracking using Python, PyTorch, and OpenCV.
- Designed a companion React.js dashboard that dynamically visualizes eye movement data and model predictions, enhancing user insights.
- Integrated analytical libraries (NumPy, Pandas, Scikit-Learn) to facilitate detailed reporting and performance tracking of the model.
- Technologies: Python, PyTorch, OpenCV, NumPy, Pandas, Scikit-Learn

Healthcare Data Management System

- Designed a semantic web-based system using ontology and RDF triples to facilitate complex healthcare data queries.
- Developed a React.js front-end that provided an intuitive interface for querying and visualizing healthcare data, enabling efficient data retrieval and analysis.
- Integrated SPARQL endpoints to support dynamic data fetching and real-time updates on the front end.
- Conducted performance testing and optimization to ensure the application could handle large volumes of healthcare data seamlessly.
- Technologies: React.js, SPARQL, Protégé, Stardog

EDUCATION

Bachelor of Science in Computer Science

July 2016 - June 2020

Vellore Institute of Technology, Vellore